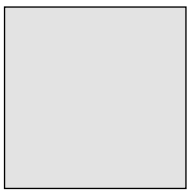
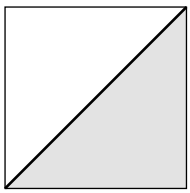
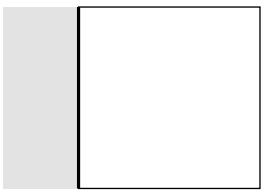
Whole squares are all gray. A cover uses every square with some gray in it. A touch at an edge or corner adds no square.



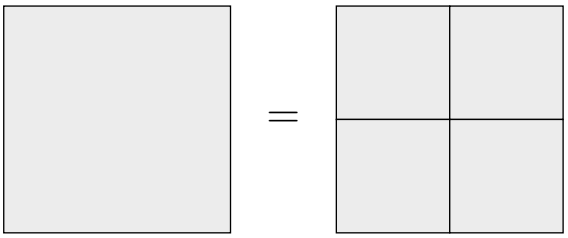
whole



partial

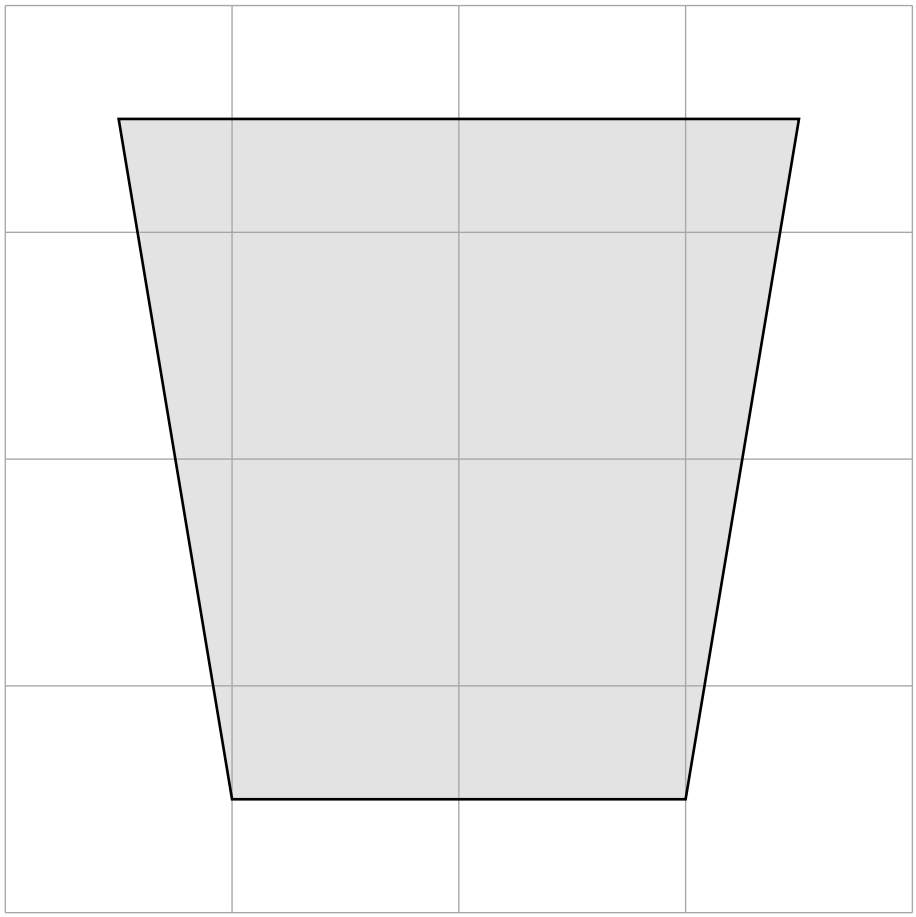


outside



1 big square
= 4 small squares

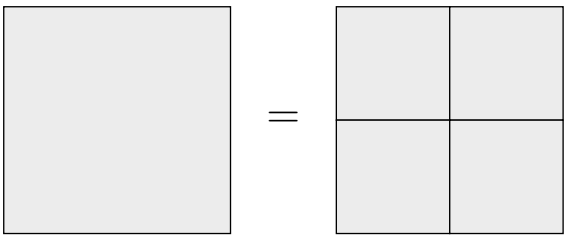
Problem 5: Split exactly three big squares into four small squares each. Choose the three that most reduce the gap between your inside and outside bounds.



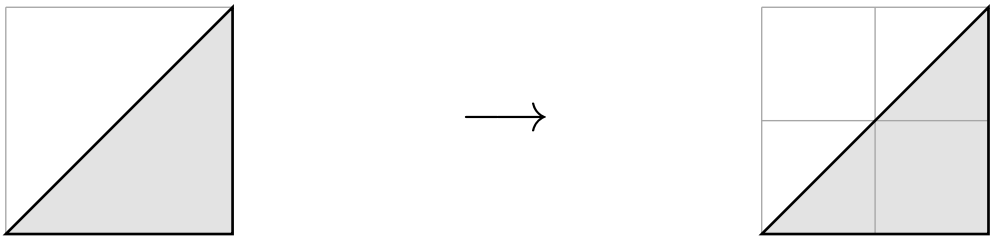
Inside: _____ big-square units

Cover: _____ big-square units

Problem 6: A big square is split into four small squares. Could the total inside area go down? Could the cover's area go up? Explain why your answers work for any of these shapes.



1 big square
= 4 small squares



Problem 7: Find the exact area of the gray shape in Problem 5. Explain how it fits between the bounds you found there.

Exact area: _____ big-square units